

Simple Harmonic #3

1. Eddie wants to find the spring constant of a spring, so he hangs the spring vertically and attaches a 0.40-kg mass to the spring's end. If the spring stretches **3.0 cm** from its equilibrium position, what is the spring constant?
2. A spring with a spring constant of 30.0 N/m is attached to a 2.3 kg mass, and the system is set in motion. Find the period.
3. What mass must you attach to a spring of constant $k = 103 \text{ N/m}$ in order for it to make 20 complete vibrations in 4.0 s?
4. In the dark of night you are assigned the dubious task of determining the height of a tower upon which you will lay siege. Conveniently, a pendulum is hanging from the tower to the ground. If the pendulum has a period of 24 s, how tall is the tower?

